Problem 8

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Problem 1 *The function f is defined by* $f(x) = \dfrac{\sqrt{2x^2+1}}{3x-5}$.

(a) *Find all vertical asymptotes of f. **EXPLAIN** and ustify your answer by using appropriate limits.*

Explanation. Candidate for vertical asymptotes: $3x - 5 = 0 \implies x = 5/3$.

Test of candidate $x = 5/3$:

$$\lim_{x \to \frac{5}{3}^+} \frac{\sqrt{2x^2+1}}{3x-5} = \infty$$

Note: the limit is of the form $\dfrac{\#}{0}$, the numerator positive, and the denominator
$3x - 5 = 3(x - 5/3)$ is positive, since $x > 5/3$.

The line $x = \dfrac{5}{3}$ is a vertical asymptote of f since $\lim\limits_{x \to \frac{5}{3}^+} f(x) = \infty$.

(b) *Find all horizontal asymptotes of f. **EXPLAIN** and justify your answer by using appropriate limits..*

Explanation. Check End Behavior as $x \to \infty$:

$\lim\limits_{x \to \infty} \dfrac{\sqrt{2x^2+1}}{3x-5}$ is of the form: $\dfrac{\infty}{\infty}$

$$\begin{aligned}
\lim_{x \to \infty} \frac{\sqrt{2x^2+1}}{3x-5} &= \lim_{x \to \infty} \frac{\sqrt{2x^2+1}}{3x-5} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} \\
&= \lim_{x \to \infty} \frac{\frac{\sqrt{2x^2+1}}{\sqrt{x^2}}}{3 - \frac{5}{x}} \\
&= \lim_{x \to \infty} \frac{\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}} \\
&= \frac{\sqrt{2}}{3}
\end{aligned}$$

The line $y = \frac{\sqrt{2}}{3}$ is a horizontal asymptote of f since $\lim_{x \to \infty} f(x) = \frac{\sqrt{2}}{3}$.
Check End Behavior as $x \to -\infty$:

$\lim_{x \to -\infty} \frac{\sqrt{2x^2+1}}{3x-5}$ is of the form: $\frac{\infty}{\infty}$

$$\begin{aligned}
\lim_{x \to -\infty} \frac{\sqrt{2x^2+1}}{3x-5} &= \lim_{x \to -\infty} \frac{\sqrt{2x^2+1}}{3x-5} \cdot \frac{\frac{1}{-x}}{\frac{1}{-x}} \\
&= \lim_{x \to -\infty} \frac{\frac{\sqrt{2x^2+1}}{\sqrt{x^2}}}{-3+\frac{5}{x}} \\
&= \lim_{x \to -\infty} \frac{\sqrt{2+\frac{1}{x^2}}}{-3+\frac{5}{x}} \\
&= \frac{-\sqrt{2}}{3}
\end{aligned}$$

The line $y = -\frac{\sqrt{2}}{3}$ is a horizontal asymptote of f since $\lim_{x \to -\infty} f(x) = -\frac{\sqrt{2}}{3}$.